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RESEARCH ARTICLE

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Phylogenetic systematics of Juncaceae

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Abstract

Premise: Juncaceae has needed taxonomic revision for some time. Specifically, the genus *Juncus* s.l. is known to be paraphyletic because five small southern-hemisphere genera have been shown repeatedly to be nested within it. In 2022, a new classification was proposed, based on phylogenies built from one nuclear and three plastid regions sequenced across much of Juncaceae, meant to resolve the paraphyletic nature of *Juncus* s.l. It created six new genera and was criticized on the basis that the genera proposed are not necessarily monophyletic, given the limited nature of the phylogenetic analyses available and the fact that the generic circumscriptions draw heavily on the morphology-based classifications in the latest monograph of Juncaceae from 2002.

Methods: We used the Angiosperms 353 target capture probe set to assemble a data set consisting of hundreds of plastid and nuclear loci to assess the monophyly of the newly proposed genera of Juncaceae.

Results: The proposed genera mostly represent monophyletic groups, but the proposed genus *Juncinella* is nested within the proposed genus *Boreojuncus*. Additionally, *Juncinella capitata* was placed as sister to either *Luzula* or *Oreojuncus* and should be recognized as a monotypic genus. Finally, *Australojuncus cyperoides* and *Verojuncus chlorocephalus* were recovered outside of their morphologically assigned genera and require further investigation to be placed confidently.

Conclusions: We proposed taxonomic revisions to rectify the stated issues, but further research is necessary, particularly to correctly place the South African annual taxa in *Juncus* s.l.

KEYWORDS

Angiosperms 353, gene tree incongruence, genome skimming, Hyb-seq, Juncus, polyploidy, target capture

Juncaceae, a family with 485 species accepted by Kew Plants of the World, has a nearly cosmopolitan distribution (POWO, 2025). Species tend to be found in mesic environments, but have colonized a wide variety of habitats, including arid deserts, coastal dunes, alpine fell fields and chaparral (Kirschner, 2002). Notably, diversity in Juncaceae is concentrated at high and low latitudes and is depauperate in the tropics, where taxa are primarily found in mountainous regions. Species diversity is particularly high in the California Floristic Province and in southeastern Australia and New Zealand (Kirschner, 2002).

Plants in Juncaceae have been used for weaving and basketry across different traditions (Pearlstein et al., 2008; Kotze and Traynor, 2011; Andonova et al., 2023), and many play important roles in wetland ecology, where they may form a large part of the macrophyte biomass in salt and freshwater marshes (Greenwood and MacFarlane, 2006; Tumas et al., 2018). As with many other angiosperm groups, the family has a range of ploidy levels and a high degree of variability in diploid chromosome number, with nuclear DNA content varying 15-fold in *Luzula* DC. and chromosome number nearly 10-fold in *Juncus* L. s.l. (Záveská Drábková, 2013). Polyploidy and aneuploidy are both common, and chromosome numbers vary within and across species (Schönswetter et al., 2007; Záveská Drábková, 2013).

Some of this karyological variation is attributable to the presence of holocentric chromosomes in the genus *Luzula*. Until recently, all species of Juncaceae were considered

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likely to have holocentric chromosomes; however, recent work using immunological staining and genomic repeat analysis has revealed that several species in *Juncus* s.l. have monocentric chromosomes, and the assumption is now that all species in *Juncus* s.l. have monocentric chromosomes (Guerra et al., 2019; Mata-Sucre et al., 2023; Dias et al., 2024). Interestingly, although these species have monocentric chromosomes, there is some indication that their centromeric regions may be more diverse than found in many other monocentric chromosomes (Dias et al., 2024). This centromeric diversity may have implications for the formation of the many aneuploid series that are seen in *Juncus* s.l. (Záveská Drábková, 2013).

In contrast to the karyological variability found in the family, morphological characters tend to be relatively conserved, especially floral and inflorescence characters. Species in Juncaceae are primarily wind-pollinated and flowers typically have six bract-like tepals arranged in two whorls of three (Oriani et al., 2012). They are usually perennial and rhizomatous, but can be annual, and leaves tend to be linear, filiform, terete, or absent (Kirschner, 2002). The vast majority of species in the family have small flowers (4–8 mm in diameter) with six tepals, three or six stamens and one pistil. Tepals vary from green to brown, occasionally with red or purple features. Inflorescence structure varies from solitary to many branched and from open to congested. Leaves are generally long, linear, or filiform and can be flat to channeled, terete, or iris-like (Kirschner, 2002). Because of the lack of obvious morphological characters, especially floral and inflorescence characters, which are accepted as generally the best for delimiting phylogenetic relationships, taxonomists working on the family have often relied on minute characters such as seed coat characteristics, seed appendages, and the presence and position of bracts and bracteoles in the inflorescence to distinguish groups (Buchenau, 1890; Ertter, 1986; Balslev, 1996; Kirschner, 2002). Additionally, they have turned to leaf characteristics, which have the potential to be rapidly evolving and not indicative of deeper relationships (Buchenau, 1890; Kirschner, 2002).

Due to the difficulty in reconstructing monophyletic groups in molecular analyses of the family and the relative lack of useful morphological characters, the classification of the family has undergone several iterations. Franz Buchenau (1890, 1906) published the first monographs of Juncaceae in 1890 and 1906. He recognized five small southern hemisphere genera, *Juncus*, and *Luzula*, although he also hypothesized that the genus *Juncus* s.l. could be polyphyletic, with the South American genera *Marsippospermum* Desv. and *Rostkovia* Desv. nested within it (Buchenau, 1906; Záveská Drábková and Vlček, 2009). He further hypothesized that the primary subgeneric groups within *Juncus* are defined by the presence or absence of a pair of bracteoles, and he formally named eight subgenera based on similarities in inflorescence structure, floral structure, and leaf anatomy (Buchenau, 1890). From that time until 1990, Buchenau's nonhierarchical groupings were generally followed. In 1990, Novikov published a revision of the supraspecific taxonomy of *Juncus* and formally recognized two subgenera defined by the presence or absence of bracteoles, as well as eight sections generally agreeing with Buchenau's original groupings. The most recent morphological revision was published by Kirschner, Novara, Novikov, Snogerup and Kaplan, 1999 to clarify the nomenclature of Juncaceae before publishing a monograph of the family in the World Flora. This work generally followed Buchenau's original taxonomy, but formally recognized two subgenera as done by Novikov (1990), and 10 sections have been validly published at different times. This treatment added one section, *Iridifolii*, and also recognized section *Caespitosi*, a group that Buchenau placed in the annual Graminifolii. Kirschner et al. (2002) acknowledged that section *Caespitosi* is a grouping of convenience to some degree and recognized that the relationships between the annual and perennial species included in section *Caespitosi* and *Graminifolii*, respectively, are unclear (Kirschner, 2002).

The five small southern hemisphere (SSH) genera originally described by Buchenau are united by having a solitary inflorescence. Buchenau did not form a hypothesis for the placement of the South American cushion plants *Distichia* Nees & Meyen, *Patosia* Buchenau, and *Oxychloë* Phil, but he hypothesized that *Rostkovia* Desv. and *Marsippospermum* Desv. might be closely related to the group of *Juncus* species with flowers subtended by bracteoles (Buchenau, 1906).

In 1986, Barbara Ertter conducted an extensive morphological and karyotypic study of section *Caespitosi*, focused on the North American species but including South African *Caespitosi*, and *Graminifolii*. She hypothesized that the North American species in section *Caespitosi* are more closely related to the North American species in *Graminifolii* than to the South African species in *Caespitosi* based on detailed morphological analysis of seed structures. She also hypothesized that North American *Caespitosi* and *Graminifolii* shared a closest common ancestor with the SSH genera (Ertter, 1986).

Molecular evidence has revealed that Buchenau's hypothesis of a paraphyletic *Juncus* is true in part and that all five SSH genera form a clade that is nested within *Juncus*. Several analyses of nuclear and chloroplast regions have shown the SSH genera to cluster with *Juncus* section *Graminifolii*, section *Caespitosi*, and section *Juncus*, supporting Ertter's hypothesis (Roalson, 2005; Záveská Drábková et al., 2004, 2006; Do and Záveská-Drábková 2018). However, these analyses could not resolve which clades were sister to the SSH genera. They thus recommended more sampling of sections *Graminifolii* and *Caespitosi* because these species appear most likely to form the sister clade to the SSH genera.

Recently, Brožová et al. (2022) and Proćków and Záveská-Drábková (2023) proposed a new classification for the family, meant to resolve the paraphyletic nature of *Juncus* s.l. Drawing from one nuclear and three chloroplast regions that have been sequenced across much of Juncaceae, they proposed six new genera that largely aligned with the

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sectional taxonomy published by Kirschner (2002) and also with the clades recovered in most molecular phylogenetic analyses of Juncaceae (Roalson, 2005; Záveská Drábková et al., 2004, 2006). Due to ongoing phylogenetic uncertainty in the molecular trees, they retained Kirschner's circumscription of section *Caespitosi* (genus *Juncinella* sensu Brožová et al. [2022]), despite it being polyphyletic in their phylogeny. The proposed classification was criticized by Elliot et al. (2023) on the basis that the genera proposed are not necessarily monophyletic, given the limited nature of the phylogenetic analyses available and the fact that the generic circumscriptions draw heavily on the morphology-based classification of Kirschner (2002). One concern is that as new phylogenetic work using genome-scale markers is done, some of these groups will not prove to be monophyletic, necessitating further taxonomic revision. Specifically, they cited the polyphyly of *Juncinella* and concerns that the Brožová et al. (2022) study showed *Boreojuncus* to be nested within *Verojuncus*. They additionally brought up concerns about the large number of nomenclatural changes associated with adopting the new classification, given that the genera being conserved represent a very small proportion of the species in the family (Elliott et al., 2023).

Here we constructed a phylogeny of Juncaceae, with the goals of clarifying poorly resolved regions and establishing a clear backbone for the family. We used the Angiosperms 353 target capture kit and genome skimming techniques to produce orders of magnitude more sequence data than has been used in previous analyses and included 28 taxa that were not previously incorporated into a phylogenetic study. We particularly focused on obtaining southern-hemisphere taxa, which have been underrepresented in previous phylogenies, and on better representing sections *Caespitosi* and *Graminifolii*. We examined the monophyly of genera in the classification of Brožová et al. (2022) and Proćków and Záveská-Drábková (2023) and propose updates to the classification. Here we refer to the larger, paraphyletic circumscription of *Juncus* as *Juncus* s.l. and use the generic names proposed by Brožová et al. (2022) (*Juncus*, *Verojuncus*, *Alpinojuncus*, *Boreojuncus*, *Australojuncus*, *Agathyron*, and *Juncinella*) when referring to that classification system. We refer to individual taxa using names used in the classification of Brožová et al. (2022) and Proćków and Záveská-Drábková (2023) to facilitate discussion on their placement in our phylogenies in light of the proposed classification.

MATERIALS AND METHODS

Sampling

We obtained dried leaves from 48 herbarium specimens distributed across all the major hypothesized clades in *Juncus* s.l. and the five SSH genera from CANB, CHR, DAV, MEL, MO, and MW. We collected 12 samples in California and dried the fresh leaves in silica gel. We further obtained 21 samples of DNA previously extracted from herbarium specimens (see Drábková et al., 2006). Herbarium voucher information is shown in Appendix 1. We downloaded DNA sequence data from an additional 17 specimens from the European Nucleotide Archive. This sampling effort represents about 25% of the species diversity in the clade containing *Juncus* s.l. and the five SSH genera. Using the taxonomy of Kirschner (2002) and Záveská Drábková (2010), we selected species to proportionately sample all hypothesized clades to minimize phylogenetic error due to biased sampling.

DNA extraction, library preparation, target enrichment, and sequencing

We froze about 20 mg of dried leaf tissue from herbarium specimens and silica-dried fresh specimens in liquid nitrogen and ground them using a mortar and pestle or a D1030 BeadBug bead mill (Benchmark Scientific, Sayreville, NJ, USA). Genomic DNA was extracted from the ground tissue using the DNeasy Plant Mini Kit (Qiagen, Hilden, Germany) and the manufacturer's protocol or the CTAB method based on Drábková et al. (2002). We quantified DNA concentration using a Qubit Fluorometer (dsDNA HS Assay Kit, Invitrogen, Foster City, CA, USA). We used the NEBNext Ultra II FS DNA library prep kit for Illumina with NEBNext Multiplex Oligos (New England Biolabs, Beverly, MA, USA) to prepare dual-indexed DNA libraries for multiplexed sequencing, using the manufacturer's protocols, except that we reduced reaction volumes by half. All samples were fragmented enzymatically in the first stage of library preparation. Insert size was set at 250 bp for all libraries, and the PCR was run for six cycles using the time and temperatures specified in the protocol. We quantified DNA concentrations for prepared libraries using Qubit and assessed size distributions using 1% agarose electrophoresis. Libraries were pooled by concentration into eight pools before the capture reaction. We used the Angiosperm 353 target enrichment kit from MyBaits (Daicel Arbor Biosciences, Ann Arbor, MI, USA) and the manufacturer's protocol to enrich the indexed libraries, with the exception of creating pools of eight libraries each per hybridization reaction. The amplification step in the PRC was run with 20–22 cycles depending on starting concentration. Post enrichment, we added 0.5 μL of unenriched libraries to ensure adequate off-target sequence capture. Enriched libraries were multiplexed and sequenced on an Illumina Hi-seq. 6000 (San Diego, CA, USA) to produce 2 × 150-bp paired-end reads. The expected number of reads per sample was ~1 million.

Sequence analysis and filtering

We used Trimmomatic to trim the reads of the sequencing output, removing adapters and low-quality reads (leading 3, trailing 3, minlength 36) and FastQC (Andrews, 2010) for a quality control check. Initial number of reads recovered per sample ranged between 1248 and 2,174,662 with a mean of 456,939. We excluded eight samples for which we recovered

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less than 100,000 reads to avoid issues due to low coverage. The demultiplexed and trimmed libraries are available under GenBank Project PRJNA1196610. We then combined our sequences with those downloaded from the European Nucleotide Archive and used the Hyb-Seq pipeline HybPiper version 2.1.6 (Johnson et al., 2016) to extract and map gene sequences. We generated a target file by filtering the mega 353 targets file provided at https://github.com/chrisjackson-pellicle/NewTargets to retain only sequences from taxa in the Poales. We mapped and aligned sequences using BLASTx (Camacho et al., 2009), known to result in longer sequences (Murphy et al., 2020). We assembled loci with SPAdes (Bankevich et al., 2012) and extracted gene sequences (Johnson et al., 2016). In addition to retrieving the default output of HybPiper, which consists of assembled protein coding sequences (exons), we extracted flanking and noncoding intron and intergenic sequences with the HybPiper script interonerate.py.

To extract plastid sequences produced as a beneficial side effect of off-target reads, we used similar methods, but we used a target file composed of the 87 coding sequences present in the *Juncus effusus* chloroplast genome (GenBank accession MW366789.1). To extract plastid data for the 19 species in Cyperaceae that we used as an outgroup, we used a target file consisting of 93 coding sequences present in the *Cyperus esculentus* chloroplast genome (GenBank accession MW542207.1). Because *Juncus* s.l. is known to have multiple ploidy levels in the genus, we used the allele phasing pipeline described by Kates et al. (2018; available at https://github.com/mossmatters/phyloscripts/tree/master/alleles_workflow) to uncover additional heterozygosity potentially due to paralogs and masked by consensus sequences (Kates et al., 2018). This pipeline maps raw reads to the supercontig consensus sequences, identifies variant sites, matches overlapping variant reads to produce separate sequences for the top two haplotypes, and aligns each haplotype across samples. We estimated gene trees from these alignments to determine whether haplotypes from individual samples group together or whether separate clades form, indicating paralogous lineages. We calculated summary statistics for the recovery using the HybPiper script hybpiper_stats.py.

Phylogenetic analysis

We produced three different data sets and analyzed them separately as follows.

Angiosperms 353 nuclear DNA

We first removed genes with less than 50% occupancy across taxa and with paralog warnings for any of the samples, then aligned genes separately using MAFFT version 7, with accuracy-oriented methods (LINSI) (Katoh and Standley, 2013). We trimmed matrices using trimAl version 1.4.rev15 with the -gappyout parameter, then created gene trees for individual genes using IQ-TREE2 version 2.2.6 (Capella-Gutiérrez et al., 2009; Minh et al., 2020). We then used Treeshrink version 1.3.7 to remove samples with excess branch lengths, indicating possible alignment errors or paralogy (Mai and Mirarab, 2018). We realigned the gene sequences subsequent to pruning, created a second round of gene trees, collapsed any nodes with less than 10% bootstrap support using newick_utils (Junier and Zdobnov, 2010), and summarized these trees using ASTRAL III (Zhang et al., 2018) to construct a multispecies coalescent tree. We then used phyparts (Smith, 2015) to assess gene tree concordance at each node. For each node, we calculated the percentage of gene trees that support the node and that conflict with it, only using gene trees that include the outgroup (*Prionium serratum*). This method has been shown to produce complementary information to node support values (Smith et al., 2015). Clade support was assessed using local posterior probability (Zhang et al., 2018). We also concatenated trimmed gene alignments and analyzed the concatenated matrix using IQ-TREE2 version 2.2.6. The matrix was initially partitioned by gene, and we used ModelFinder to merge partitions to a best fit partition scheme as well as select substitution models. Clade support was assessed using 1000 ultrafast bootstraps (Minh et al., 2020). After initial tree building, we inspected tips that grouped separately from other samples of the same taxon, and dropped accessions that appeared likely to be misidentified, or contaminated. We then constructed final trees using the methods described above.

Plastid data set

We first removed genes with less than 50% occupancy across taxa, then aligned genes separately using MAFFT version 7 and accuracy-oriented methods (LINSI) (Katoh and Standley, 2013). We trimmed matrices using trimAl version 1.4.rev15 with the -gappyout parameter, then used IQ-TREE2 version 2.2.6 to analyze the concatenated matrix (Capella-Gutiérrez et al., 2009; Minh et al., 2020). We used one partition and selected the best fit substitution model using ModelFinder (Kalyaanamoorthy et al., 2017). Clade support was assessed using 1000 ultrafast bootstraps (Minh et al., 2020). After initial tree building, we inspected tips that grouped separately from other samples of the same taxon, and dropped accessions that appeared likely to be misidentified, or contaminated. We then constructed a final tree using the methods described above.

Extended plastid data set

To expand taxon sampling and ameliorate any issues that might be caused by sparse sampling and long-branch attraction, we assembled an extended plastid data set, drawing on publicly available sequence data. We used the R package phylotaR version 1.3.0 (Bennett et al., 2018) to identify *matK*, *rbcL*, *rpoC1*, and *trnA-psbA* as the best-represented gene regions for Juncaceae available on GenBank and downloaded

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all available sequences for these genes. We combined this data set with our plastid data set. To reduce the number of tips in the phylogenetic tree, in cases where there were different accessions from the same species for different gene regions, we randomly concatenated genes from the same species. Although this method is not best practice, most accessions available were from different individuals, so including them all separately was not tractable. Sample names and gene regions available are in Appendix S1.

We again dropped genes from our plastid data set that were not present in 50% of taxa and aligned genes separately using MAFFT version 7, with accuracy-oriented methods (LINSI) (Katoh and Standley, 2013). We trimmed matrices using trimAl version 1.4.rev15 with the -automated1 parameter, then used IQ-TREE2 to analyze the concatenated matrix (Capella-Gutiérrez et al., 2009; Minh et al., 2020). We used one partition and selected the best fit substitution model using ModelFinder. Clade support was assessed using 1000 ultrafast bootstraps. We also drew 2000 random quartets to assess the phylogenetic information of the input alignment. After initial tree building, we inspected tips that grouped separately from other samples of the same taxon and dropped accessions that appeared likely to be misidentified or had a sequencing error. We then constructed a final tree using the methods described above.

All figures were plotted using R version 4.0.3 (R Core Team, 2020) and R packages ape version 5.8.1 (Paradis and Schliep, 2019), ggtree version 4.0.1 (Yu, 2020), ggplot2 version 4.0.1 (Wickham, 2016), ggimage version 0.3.4 (Yu, 2026), and their dependencies. The multiple sequence alignments used to construct the final trees for all the data sets are available at Dryad: https://doi.org/10.5061/dryad.x3ffbg810.

Gene flow analysis

To determine whether gene flow could be a significant factor leading to gene tree discordance, we used the software HyDe version 0.4.3 to detect hybridization in the presence of incomplete lineage sorting (Blischak et al., 2018). We performed an exhaustive search, testing each taxon as a possibly hybrid of all other taxon pairs and used *Prionium serratum* as the outgroup. As input for HyDe, we used the same alignment as used for the Angiosperms 353 nuclear DNA concatenated analysis.

RESULTS

Sequence recovery

We generated novel sequence data for 81 species in Juncaceae and produced orders of magnitude more data than previous sequencing efforts. On average, after trimming and dropping samples with low coverage, our sequence data set contained 803,290 read pairs per sample sequenced (minimum 142,841, maximum 2,174,662). We had variable recovery of the Angiosperms 353 genes, ranging from 24 to 347 loci, with a mean of 231 loci (Appendix 1). Surprisingly, we had the best success from samples extracted from herbarium specimens with a mean of 239 loci recovered (range 24–347) and somewhat less success with our silica-dried samples (mean 184 loci, range 72–239) (Appendix 1), perhaps because the DNeasy Plant DNA extraction kit was used for most of the silica-dried samples and modified CTAB methods were used for the herbarium specimens. We were also successful with the sequences downloaded from sequence read archives that targeted Angiosperms 353 genes, recovering a mean of 179 loci (range 0–240) (Appendix 1).

Plastid sequence recovery was similarly variable, with a mean of 46 loci out of 87 plastid coding regions recovered from the *Juncus effusus* genome, recovered across all ingroup samples. Our herbarium and silica-dried samples had similar recoveries (mean 51 loci, range 5–72 and 2–61, respectively) (Appendix 1). We had good success with sequences downloaded from sequence read archives that targeted plastid genomes, recovering a mean of 66 loci (range 61–67), but recovery was poor from sequences downloaded that targeted Angiosperm 353 genes with a mean of 17 plastid loci (range 1–61) (Appendix 1). We were able to recover the *matK*, *rbcL*, *rpoC1*, and *trnA-psbA* chloroplast markers for most samples, which we used for the extended plastid analysis (Appendix S1).

Data sets

Angiosperms 353 data set

Seventy ingroup species were included in the Angiosperms 353 data set, representing 22% of the total species diversity in *Juncus* s.l., but covering 100% of the described subgenera and sections sensu Kirschner 2002 (Table 1). This tree was rooted using 25 species in Cyperaceae and *Prionium serratum* (Thurniaceae) (Species and accession numbers for outgroup Cyperaceae species are listed in Appendix S2: Table S1.). We excluded 154 loci that were not represented in enough samples or because they were flagged as paralogs, leaving 199 loci to be analyzed (14,981 individual sequences and 127,894 sites). The alignment had 81% variable sites, 65% parsimony informative sites, and 6% missing data (Table 1). The gene alignments included sequences from 73 individuals on average (range 39–103), their average length was 252 bp (range 58–1474), and on average, 64% of sites were parsimony informative (range 31–93%) (Appendix S3).

Plastid data set

The plastid data set included 68 of the same ingroup species as the Angiosperms 353 data set, with an additional 13 species in *Luzula* added, for a total of 81 ingroup species (Table 1). This tree was rooted using 19 species in Cyperaceae and *Prionium serratum* (Thurniaceae). The alignment was composed of 59

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TABLE 1 Summary of DNA sequence recovered from species in Juncaceae for three datasets, nuclear (sequences recovered for the angiosperms 353 nuclear loci), plastid (sequences recovered for coding sequences from the Juncus effusus chloroplast genome) and plastid extended (plastid plus sequences for *matK*, *rbcL*, *rpoC1*, and *trnA-psbA* downloaded from GenBank and the European Nucleotide Archive). Summary statistics are displayed for multiple sequence alignments generated using the above datasets.

	Data set		
	Nuclear	**Plastid**	**Plastid extended**
No. of markers	199	59	59
No. of tips (total/ingroup)	107/81	113/93	730/710
No. species (ingroup)	70	81	226
No. of sequences	14981	5114	6005
No. of sites	127894	174096	53441
Variable sites (%)	81	44	66
PIS[a] (%)	65	25	39
Missing data (%)	6	22.3	87

[a]Parsimony informative sites.

concatenated loci and contained 174,096 sites, 44% of which were variable and 25% of which were parsimony informative. The alignment had 22% missing data (Table 1). On average, the locus alignments contained sequences from 86 individuals (range 60–105), and 2729 sites (range 1605–9218). On average, 25% of sites were parsimony informative (range 12–41%) (Appendix S4).

Extended plastid data set

We included 730 samples in this data set, representing 226 ingroup taxa (51% of the taxa in Juncaceae) (Table 1). There were 139 taxa in *Juncus* s.l., representing 44% of the total taxa in the genus, and 75 taxa in *Luzula*, representing 65% of the total taxa in the genus. All other genera in Juncaceae were represented, as were outgroup species in Cyperaceae and *Prionium serratum* (Thurniaceae). The concatenated alignment contained 53,441 sites; however, it had 87% missing data because we retained all of the plastid sequences from the plastid data set, but most samples were represented only by the four markers we chose to download from the sequence read archives. The alignment had 66% variable sites and 39% parsimony informative sites (Table 1). The locus alignments contained on average 91 taxa (range 60–689), and 686 sites (range 217–4650). We recovered by far the most sequences for the *rbcL* locus (689), followed by *matK* (263), *rpoC1* (189) and *trn-psbA* (149) (Appendix S5).

Phylogenetic relationships

We recovered slightly different topologies across our three data sets and analytical methods. Here we will refer to fully supported clades as having 100% ultrafast bootstrap support, or a local posterior probability of 1, well-supported clades as having higher than 90% ultrafast bootstrap support (local posterior probability > 0.9), and moderately supported clades as having between 50% and 90% ultrafast bootstrap support (local posterior probability between 0.5 and 0.8). Juncaceae was recovered as monophyletic in all trees. Across all trees, *Verojuncus chlorocephalus* came out as either the outgroup to the rest of Juncaceae, or formed a clade with *Juncinella cappillaris*. In our Angiosperms 353 tree, *Juncinella capillaris* came out as sister to *Verojuncus chlorocephalus* (Figure 1), but in our plastid trees (Figure 2), it grouped with the other species in *Juncinella*. One other species in our data set grouped with its conspecifics in the plastid trees, but not in the Angiosperms 353 tree. This sample had additional representation of the same species in our data set and grouped with other samples that had low read depths or were misidentified, so we removed it from further analysis, given that its placement seemed likely to be due to contamination or poor sequence quality. High levels of discordance among the nuclear gene trees likely exacerbated this issue. On average, across all nodes in the Angiosperms 353 tree, only 19% of the gene trees in our gene concordance analysis were concordant with a particular node (Figure 1).

The genera proposed by Brožová et al (2022) and Proćków and Záveská-Drábková (2023) were generally well supported, but there were a few key issues that were not resolved. Three of the proposed genera were fully supported across all trees and all trees recovered the same topology. *Alpinojuncus* was recovered as sister to *Verojuncus*, with full support across all trees. *Australojuncus* was recovered as sister to the clade containing *Distichia, Oxychloë* and *Patosia* with full support across all trees. However, one species (*Australojuncus cyperoides*) came out inside of *Verojuncus* in all analyses. This South American taxon came out within a clade of Australian *Verojuncus*.

Agathyron was fully supported as monophyletic across all trees, but slightly different topologies were recovered across trees. Our plastid and extended plastid trees recovered *Agathyron* as sister to all other *Juncus* s.l. as well as the SSH genera (Figure 2), whereas our Angiosperms 353 tree recovered *Agathyron* as sister to *Verojuncus* and *Alpinojuncus* (Figure 1).

Juncus was fully supported in our plastid and Angiosperms 353 trees and came out as sister to the SSH genera, *Boreojuncus*, *Australojuncus*, and *Juncinella*. However, in our extended plastid analysis, *Juncus krausii* was recovered as the outgroup to *Juncus*, *Boreojuncus*, the SSH genera, *Australojuncus*, and *Juncinella* (Appendix S2: Figure S3A–C).

Boreojuncus and *Juncinella* were well supported across all trees, but across all trees *Juncinella* was nested within *Boreojuncus*. In the plastid and Angiosperms 353 data sets (Figures 1, 2), this topology was only due to the placement of *Boreojuncus marginatus*, but in the extended plastid tree, *Boreojuncus repens* came out with *B. marginatus*, confirming the nested status of *Juncinella* (Appendix S2: Figure S3A–C). *Juncinella* is additionally problematic in that *Juncinella*

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COLOR FIG

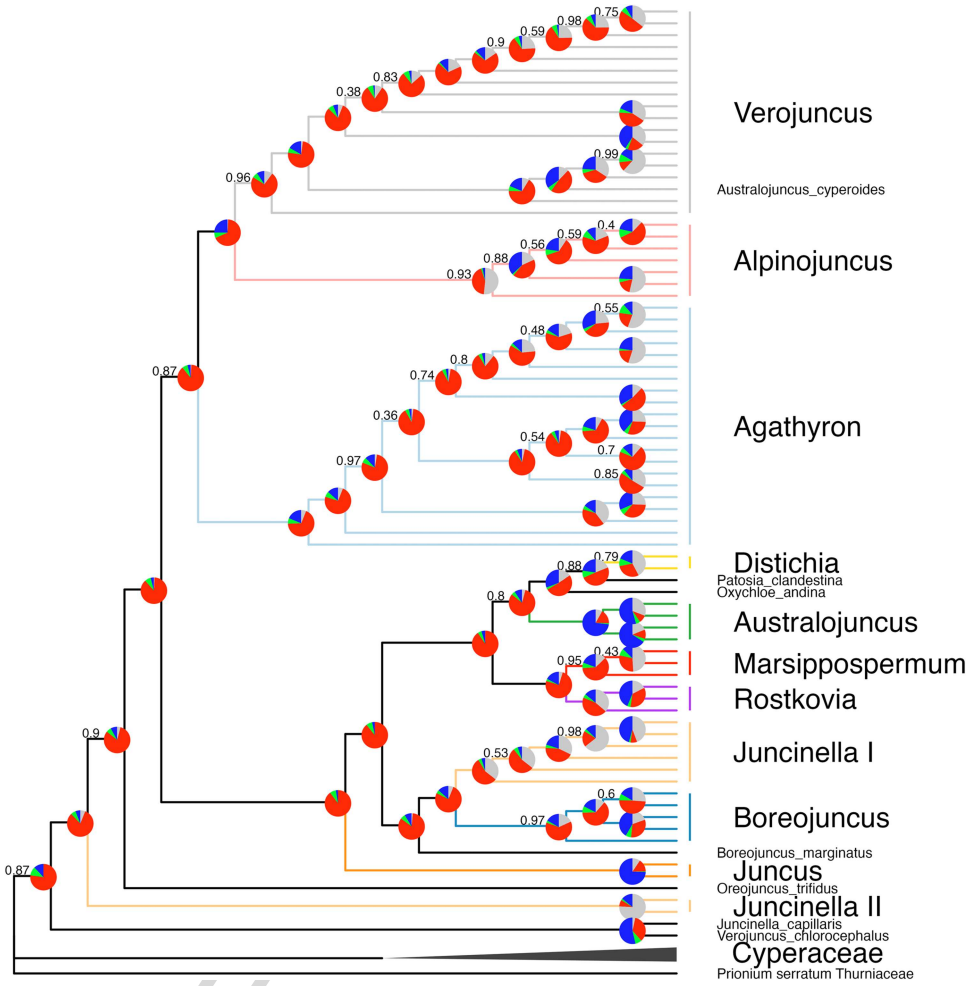

FIGURE 1 Phylogenetic reconstruction of Juncaceae based on the Angiosperms 353 nuclear genes and generated using ASTRAL III. Outgroups are collapsed; clades are colored by focal genera. Pie charts represent the percentage of genes in the gene concordance analysis that support the node (blue), the proportion that support the main alternative for that node (green), the proportion that support the remaining alternatives (red), and the proportion that inform (support or conflict) this clade that have less than 50% bootstrap support (grey). Numbers above nodes are local posterior probability, only nodes with local posterior probability less than 1 are labeled.

capitata came out as sister to *Luzula*, or sister to *Oreojuncus*, indicating that it should not be included in *Juncinella*.

The five genera that were previously nested within *Juncus* s.l. formed a clade that includes *Australojuncus*. *Rostkovia* and *Marsippospermum* are sister genera and formed a well-supported clade across all trees. *Distichia*, *Patosia*, and *Oxychloë* formed a well-supported clade, but the genera were only recovered as monophyletic in the Angiosperms 353 tree (Figure 1); otherwise, species occupied varying positions in the clade. *Oreojuncus* was recovered as external to the rest of *Juncus* s.l. and the SSH genera across all trees, and in the extended plastid tree, it formed a clade with *Luzula* and *Juncinella capitata*. *Luzula* was recovered as monophyletic across all trees, and came out as sister to *Juncus* s.l.

Gene flow analysis

We conducted 171,465 three-taxon comparisons using HyDe. Of these, 48,135 (28%) detected significant evidence of hybridization. Taxon triplets with significant scores included many where the tested species were from different genera, indicating that many of them may be reflecting ancestral hybridization events, as opposed to current day introgression. Because this study was not focused on mapping ancestral hybridization events in the family, we interpreted our results as supporting gene flow as a likely factor contributing to gene tree discordance and did not perform any further analyses.

DISCUSSION

Here we present the most comprehensive phylogenetic analysis of Juncaceae to date, with 51% of all species in the family represented. The analyses include large numbers of nuclear and plastid markers and fully resolve most clades in the family. The results firmly establish the phylogenetic position of the SSH genera and included 36 species that were not previously placed in a phylogenetic context.

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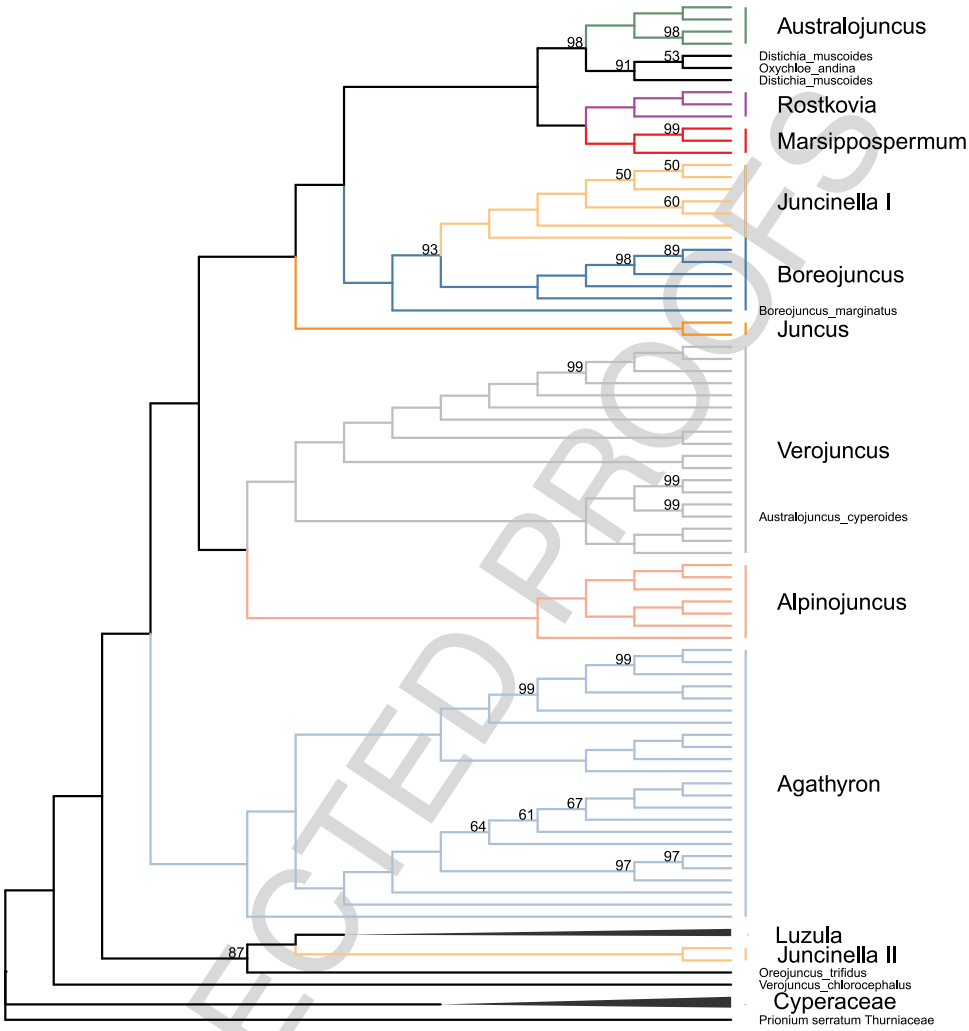

FIGURE 2 Phylogenetic reconstruction of Juncaceae based on plastid loci and generated using IQ-TREE2. Outgroups are collapsed; clades are colored by focal genera. Numbers above nodes are ultrafast bootstrap support values. Only values less than 100 are shown.

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Our analysis of 353 nuclear loci resulted in well-defined clades with high support values and a good ability to resolve relationships to the species level. This analysis also revealed extreme levels of gene tree discordance between nuclear genes, and our HyDe analysis suggests that there could be a strong signature of ancestral hybridization in the nuclear genome for this family. The high levels of discordance between nuclear genes likely led to some instability when placing particular taxa. *Juncinella capillaris*, for example, is placed with its congeners in our plastid analysis, but with *Verojuncus chlorocephalus* in our final Angiosperms 353 analysis. In some variations of our Angiosperms 353 analysis, such as when using a different target file, *Juncinella capillaris* is placed with the rest of *Juncinella*. *Juncinella capillaris* is very clearly placed with other species in *Juncinella* with respect to morphology and geography. It is morphologically very distinct from *Verojuncus chlorocephalus*, to the degree that any hybridization between the two taxa would almost certainly be evident in the morphology of the specimen. *Verojuncus chlorocephalus* is placed as the outgroup to all of the Juncaceae in the Angiosperms 353 analysis, and samples with poor sequence quality can often be placed near the outgroup due to long-branch attraction. Given these facts, we hypothesize that the odd placement of *Juncinella capillaris* in the Angiosperms 353 tree is more likely to be due to sequence quality or slight contamination, rather than a biological connection to *Verojuncus chlorocephalus*. One other sample showed similar a pattern, where it grouped with conspecifics in the plastid analysis, but not in the Angiosperms 353 analysis. This sample seemed to have a similar problem as *Juncinella capillaris*, where it was placed with samples that we dropped due to low read coverage after our initial analyses. This sample had additional samples of the same species present in the data set, so we adopted a conservative approach and excluded it from further analysis. The Angiosperms 353 loci proved to be a rich source of phylogenetic information for this family, with potential to reveal insight into more complex evolutionary processes such as introgression. However, our results indicate that the analysis of these data

may be more sensitive to sequence quality than for the plastid data. This finding is relevant to projects using a variety of sources for genetic material such as herbarium specimens of different ages and qualities. Of note is the fact that CTAB-based extraction methods may be more effective for obtaining high-quality sequence data for Juncaceae than conventional kit methods; however, sequence quality can still be an issue.

Our analyses using plastid markers were also effective at recovering well-supported clades that are congruent with morphology and geography, at least at the generic level. The plastid analyses of our core data set had a slightly lower ability to resolve species-level relationships than the analysis using nuclear loci, and the extended plastid analysis was poor at resolving species-level relationships, as we would expect. However, the plastid analysis recovered well-supported clades deeper in the phylogeny, that were largely congruent with the clades recovered using the nuclear loci. Though the position of *Agathyron* is still undetermined, plastid data still represent a rich source of phylogenetic information, and analyses with large numbers of plastid loci proved to be effective for resolving most relationships between genera in Juncaceae. However, a broader taxon sampling for complete plastid genomes would be highly beneficial, given the high degree of missing data in our extended plastid data set. Determining the correct placement of *Agathyron* will likely need both nuclear and chloroplast data, and incongruence between our chloroplast and nuclear data at this node suggests a complex evolutionary history at the deep nodes of the Juncaceae. However, for the purposes of placing taxa into genera in the family, plastid data has great utility, although many plastid loci likely need to be sampled to establish high confidence in placements. Although plastid data only represents part of the evolutionary history, our analyses suggest that it is effective at recovering genus-level clades and may be more approachable for studies that need to use genetic material from a variety of sources. Likely because the plastid data lack some of the complexity of nuclear data and because loci are present at high copy numbers, we saw that plastid data was less sensitive to issues with poor sequence quality.

Because the plastid does not experience the same effects from polyploidy, incomplete lineage sorting, and introgression as the nuclear genome, it can at times be more reliable for reconstructing the history of speciation (Xiong et al., 2009; Zhang et al., 2017; Zeb et al., 2020). Although plastid data can also be associated with a lack of ability to resolve fine-scale phylogenetic relationships, we found the plastid regions to be effective at recovering generic level relationships across the family phylogeny. We recommend this strategy for future studies to assess intermediate level relationships across the family. Studies addressing fine-scale relationships between species likely require nuclear sequence data and could benefit from techniques such as family specific probe design to minimize paralog capture (Moore-Pollard et al., 2024). More explicit analysis of hybridization between lineages is also likely to be fruitful when looking at species-level relationships in this family, such as variants of the *D* statistic or inference of phylogenetic networks (Durand et al., 2011; Patterson et al., 2012; Wen et al., 2018).

Though the evolutionary complexity in the nuclear genome of Juncaceae can present analytical challenges, it also indicates an interesting evolutionary history. We saw a large degree of gene tree discordance in our Angiosperms 353 data set, especially at deeper nodes in the phylogeny (Figure 1). Although we excluded any genes that were flagged as potentially paralogous in our pipeline, the many ploidy levels in Juncaceae may be one reason that we saw such a high degree of gene tree discordance in the Angiosperms 353 data set. In addition to conventional issues with paralogs, in lineages with high rates of polyploidy, there is the possibility that different gene copies will be sorted into different lineages. Thus, in some lineages, putative orthologs would in reality be paralogs, but would be extremely difficult to distinguish (Maddison, 1997). Perhaps due to the extreme karyotypic complexity present in the family, our pipeline may not have filtered out all paralogs present.

Introgression is another potential driver of gene tree discordance (Morales-Briones et al., 2021). We expect our multispecies coalescent analysis to account for incomplete lineage sorting among nuclear genes, but it does not account for the potential effect of hybridization (Zhang et al., 2018). Our HyDe analysis indicates that hybridization could be common across the phylogeny, and interspecific hybridization is hypothesized to be widespread in the Juncaceae (Kirschner, 2002). We observed a pattern of increasing gene discordance at nodes deeper in the phylogeny. We did not see that any nodes had a well-supported alternative topology, rather there were many alternative topologies with very low levels of support. For this reason, we did not explore alternative splits because it was largely impossible to determine which alternative topologies would represent, for example, an ancestral hybridization event that was important for the establishment of a new lineage. The pattern we observed perhaps indicates that extensive polyploidization and frequent hybridization between lineages have been ongoing processes in the evolution of the family.

We found that the genera proposed by Brožová et al. (2022) and Proćków and Záveská-Drábková (2023) were well supported phylogenetically for the most part, with only a few discrepancies that need to be addressed. The placement of three species in our phylogenies did not agree with their morphological assignments, and we find issues or potential issues with three of the proposed genera.

Verojuncus chlorocephalus and *Australojuncus cyperoides* are both placed on our phylogeny in disagreement with their morphological groups; however, both are also represented by only one sample. *Verojuncus chlorocephalus* is native to the California Floristic Province and is distinct from many other *Juncus* s.l. in having white to pink tepals. Its perfectly septate, unitubulose leaves place this species in the genus *Verojuncus*. However, the pale tepals are

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reminiscent of many species in *Alpinojuncus*, although this group is centered in the Sino-Himalayan region, making it biogeographically implausible that *V. chlorocephalus* is closely related to this group. Our analyses consistently placed our sample of *V. chlorocephalus* as the outgroup to *Juncus* s.l., the SSH genera, and *Luzula*. However, there are some unusual characteristics of this sample. We recovered the highest number of Angiosperms 353 genes for this sample (347), leading us to question whether there could be some issue with contamination or polyploidy. In the plastid analysis, we had rather low recovery of gene regions for this sample (18), but still recovered the same topology. For this reason, we believe that this taxon requires more sampling, but may represent a rather distinct evolutionary lineage.

In all of our analyses, *Australojuncus cyperoides* is placed in a clade of Australian species in the proposed genus *Verojuncus*. Morphologically, *Australojuncus cyperoides* has flattened grass-like leaves, placing it in Kirschner's section *Graminifolii* and in the proposed genus *Australojuncus* based on its leaf morphology and its distribution in South America, although it has cauline leaves, distinct from the rest of *Australojuncus*). The specimen that we sequenced was collected in California from a small, naturalized population, so the chance that it is, in reality, a misidentified taxon from Australia is negligible. Its DNA was extracted in a separate batch from any of the Australian accessions that it groups with and only processed in the same batch as one of the accessions it groups with (*J. sandwithii*, MEL 2460301). We thus believe that the chance this placement is due to a laboratory error is minimal. The odd placement of this taxon should be confirmed by additional analyses, because its morphology and distribution clearly place it in *Australojuncus*.

The proposed genus *Juncus* is generally well supported by our analyses. The only instance of discrepancy is in our extended plastid tree, where *Juncus krausii* is placed as the outgroup to *Juncus*, *Australojuncus*, *Boreojuncus*, the SSH genera, and *Juncinella* (Appendix S2: Figure S3A–C). Due to the high levels of missing data in the extended plastid analysis, we do not place any large weight on this topology and defer any conclusions until more complete analyses.

We find more compelling evidence that *Boreojuncus* is paraphyletic and *Juncinella* is nested within it. *Juncinella* is nested within *Boreojuncus* in all of our analyses, and this pattern is well supported. This finding actually aligns with Buchenau's original taxonomy, where section *Caespitosi* sensu Kirschner (2002) are annual members of Buchenau's section *Graminifolii* (Buchenau, 1906). Geographically, it is logical that the *Juncinella* in western North America might be more related to the *Boreojuncus* found in western North America than the *Boreojuncus* in eastern North America (*Boreojuncus repens*) or South America (*Boreojuncus marginatus*). (See Appendix S6 for distribution maps for all species.) We believe we have substantial evidence to support the broader circumscription of *Boreojuncus* to include the annual species in western North America that have been classified as *Juncinella*.

Juncinella is additionally problematic, because *Juncinella capitata* was sister to either *Luzula* or *Oreojuncus* in all of our analyses, which mirrors results from other studies (Do and Záveská-Drábková, 2018; Brožová et al., 2022, Mata-Sucre et al., 2023) and creates a polyphyletic *Juncinella*. Geographically, *J. capitata* is disjunct from the western North American species that make up the majority of *Juncinella*, with its center of origin in Eurasia. Kirschner (2002) acknowledged the likely nonmonophyly of his section *Caespitosi*, with the disjunct distribution of *J. capitata* as one of the reasons (Kirschner, 2002). *Juncinella capitata* is also morphologically separated from the other species in *Juncinella*, having an herbaceous inflorescence bract, and outer tepals that are acuminate, and conspicuously longer than the inner (Kirschner, 2002). Combined with the previous morphological, geographic, and genetic works cited above, our results provide compelling evidence that *J. capitata* likely should be treated as a monotypic genus.

Juncinella, as circumscribed by Proćków and Záveská-Drábková (2023), also includes South African annual species. There have been few collections made of these species, so they are not represented in our data set, but they have been hypothesized by Ertter 1986 to be a separate clade from the North American section *Caespitosi* (*Juncinella* sensu Proćków and Záveská-Drábková [2023]), and likely most closely related to the South African grass-leaved *Juncus* (*Australojuncus* sensu Proćków and Záveská-Drábková [2023]) (Ertter, 1986). Given that our results resolve *Australojuncus* as a separate clade from *Boreojuncus*, further analysis likely will show the South African *Juncinella* to be a separate clade as well. We propose a new genus for the South African *Juncinella*, provisional upon continued support from broader sampling and additional consistency across data sets. Table 2 lists the genera proposed by Proćków and Záveská-Drábková (2023) and our newly proposed genus and the defining morphological traits of each genus (Proćków and Záveská-Drábková, 2023).

CONCLUSIONS

In our analysis of nuclear and plastid regions across Juncaceae, the genera recognized in Brožová et al. (2022) proposed classification correspond, for the most part, to groups that are well supported as monophyletic, with several exceptions. Three species do not group with the genus to which they would be assigned based on their morphology. For two of these, *Verojuncus chlorocephalus* and *Australojuncus cyperoides*, additional molecular data are needed to confirm or refute their placement. The third, *Juncinella capitata* is clearly placed outside of *Juncus* s.l. and requires a new generic name. We further found that the North American species in *Juncinella* should be included within *Boreojuncus*. Thus, we conclude that a few modifications to Brožová and Záveská Drábková's proposed classification are needed to ensure that all genera recognized are well-supported monophyletic groups. In Appendix 2, we have

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TABLE 2 List of genera in Juncaceae and their defining morphological features.

Genus	Defining features
Oreojuncus	Leaves flat to slightly canaliculate, auricles lacerate; inflorescence terminal.
Luzula	Leaves flat, grass-like, with hairs on the leaf margin; seeds 3.
Agathyron	Leaves terete, to flat or laterally compressed; inflorescences cymose, lower bract occasionally apparently forming a prolongation of stem.
Juncus	Leaves terete, leaf tip sclerenchymatous; lower bract apparently forming a prolongation of stem, flowers in head.
Verojuncus	Leaves unitubulose or pluritubulose, perfectly septate; if pluritubulose and imperfectly septate then terete, flattened or laterally compressed; inflorescence racemose.
Juncinella	Inflorescence bracts leaf-like, exceeding inflorescence, outer tepals much longer than inner, acuminate. Annuals.
Alpinojuncus	Anthers distinctly to at least partly exserted from the perianth in later stages of flowering, or if not exserted, then seed with two distinct appendages. Sino-Himalaya region.
Boreojuncus	Leaves flat, grass-like if plants perennial with 1–6 (or more) cauline leaves, or if plants annual, leaves basal, canaliculate. Western North America.
Australojuncus	Leaves flat, grass-like, plants perennial, without cauline leaves; if cauline leaves present, then *Australojuncus cyperoides*. Native only in southern hemisphere.
Australojuncinella	Annuals. Leaves flat, grass-like, none cauline. Native only in southern hemisphere.
Patosia	Leaf margin minutely serrulate; inflorescences solitary; plants mat-forming.
Distichia	Leaves terete, distichous; inflorescence solitary; plants mat-forming.
Oxychloë	Leaves terete, spirally arranged; inflorescence solitary; plants mat-forming.
Rostkovia	Leaves linear or with reduced blade, all basal; inflorescence solitary, flower bracts 2, with the longer one erect to suberect, much exceeding the flower.
Marsippospermum	Leaves terete, cauline; inflorescence solitary, flower bracts short, membranous.

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elevated *Juncinella capitata* to a monotypic genus, updated the circumscription of *Boreojuncus* to include North American species formerly in *Juncinella*, and created a new genus *Australojuncinella* that includes the South African species formerly in *Juncinella*.

Our analyses resolved the majority of the longstanding questions about the backbone phylogeny of Juncaceae, especially as they pertain to the exact placement of the SSH genera. However, we also uncovered additional questions, such as the proper placement of *Verojuncus chlorocephalus* and whether the unexpected placement of *Australojuncus cyperoides* might indicate that *Australojuncus* could still be polyphyletic. We were also not able to obtain any tissue samples representing the South African species in *Juncinella* (our proposed genus *Australojuncinella*) and believe that resolving the placement of these species relative to *Australojuncus* should be a priority.

AUTHOR CONTRIBUTIONS

R.K.: conceptualization (lead); data curation (lead); formal analysis (lead); writing original draft (lead); and editing (equal); funding acquisition (equal); investigation (lead); methodology (lead); resources (equal); visualization (lead). L.Z.D.: conceptualization; resources (equal); review and editing (equal). D.P.: funding acquisition (equal), supervision (lead), review and editing (equal).

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CONFLICT OF INTEREST STATEMENT

Daniel Potter is an Associate Editor of the *American Journal of Botany* but took no part in the peer-review and decision-making processes for this paper.

DATA AVAILABILITY STATEMENT

Demultiplexed and trimmed libraries are available under GenBank Project PRJNA1196610. Multiple sequence alignments can be downloaded at Dryad (https://doi.org/10.5061/dryad.x3ffbg810). Voucher and accession numbers are in Appendix 1.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. List of samples in extended plastid tree and GenBank markers used.

Appendix S2. Supplemental trees with branch lengths and full species labels, trees labeled with our new proposed groupings and showing geographic distributions, and list of species and accession numbers used as outgroups.

Appendix S3. Gene alignment statistics for nuclear regions.

Appendix S4. Gene alignment statistics for plastid regions.

Appendix S5. Gene alignment statistics for extended plastid regions.

Appendix S6. Range maps for all species present in analysis.

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Appendix 1. Summary of sample sources and recovery. Herbaria codes are from the Index Herbariorum (Thiers, 2025). Taxon names are as found on herbarium record.

Herbarium code	Taxon	Voucher	Source: Accession	353 loci	Plastid loci
AAU	*Juncus pygmaeus*	AAU J. E. de Laghe 2/23/2005	Herbarium	225	41
B	*Marsippospermum philippii*	B M. Weigend, H. Forther, N. Dostert and K. Weigend 6053 (ref. 2002/244)	Herbarium	293	51
BM	*Juncus maritimus*	BM L. Holm-Nielsen, B. Ollgaard and S. P. Pinerup 807	Herbarium	303	41
CANB	*Juncus continuus*	CANB 691882	Herbarium	162	68
CANB	*Juncus acutus subsp. acutus*	CANB 821884	Herbarium	295	16
CANB	*Juncus prismatocarpus*	CANB 823903	Herbarium	218	61
CANB	*Juncus homolocaulis*	CANB 824865	Herbarium	282	32
CANB	*Juncus sandwithii*	CANB 825241	Herbarium	86	1
CANB	*Juncus durus*	CANB 85747	Herbarium	220	25
CANB	*Juncus holoshoenus*	CANB 878610	Herbarium	246	25
CANB	*Juncus fockei*	CANB 896358	Herbarium	229	59
CANB	*Juncus falcatus*	CANB 909667	Herbarium	295	60
CANB	*Juncus planifolius*	CANB 910211	Herbarium	287	54
CANB	*Juncus caespiticius*	CANB 918420	Herbarium	271	53
CHR	*Juncus pusillus*	CHR 365317	Herbarium	271	57
CHR	*Juncus caespiticius*	CHR 417040	Herbarium	218	35
CHR	*Juncus scheuchzerioides*	CHR 639436	Herbarium	224	58
CHR	*Rostkovia. magellanica*	CHR272165	Herbarium	147	18
CHR	*Juncus scheuchzerioides*	CHR639494	Herbarium	135	5
CHR	*Rostkovia magellanica*	CHR667324	Herbarium	286	67
CHR	*Rostkovia magellanica*	CHR688848	Herbarium	264	47
MO	*Oxychloë andina*	D. motesino 2291	Herbarium	214	15
DAV	*Juncus breweri*	DAV 334599	Herbarium	250	27
DAV	*Juncus chlorocephalus*	DAV 334915	Herbarium	347	18
DAV	*Juncus tenuiss*	DAV 334964	Herbarium	295	47
DAV	*Juncus occidentalis*	DAV 343406	Herbarium	296	46
DAV	*Juncus parryii*	DAV 359706	Herbarium	296	41
DAV	*Juncus macrophyllus*	DAV237186	Herbarium	205	71
DAV	*Juncus covillei*	DAV237189	Herbarium	290	72
DAV	*Juncus tiehmii*	DAV237191	Herbarium	111	63
DAV	*Juncus capitatus*	DAV237195	Herbarium	258	60
DAV	*Juncus mertensianus*	DAV237197	Herbarium	227	59
DAV	*Juncus orthophyllus*	DAV237198	Herbarium	249	70
DAV	*Juncus howelii*	DAV237199	Herbarium	243	65
DAV	*Juncus capillaris*	DAV237201	Herbarium	339	60
DAV	*Juncus luciensis*	DAV237202	Herbarium	236	61
DAV	*Juncus bulbosus*	DAV334591	Herbarium	191	55

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Herbarium code	Taxon	Voucher	Source: Accession	353 loci	Plastid loci
DAV	*Juncus bufonius*	DAV334800	Herbarium	275	67
DAV	*Juncus cyperoides*	DAV334866	Herbarium	316	47
DAV	*Juncus tenuiss*	DAV334982	Herbarium	231	71
DAV	*Juncus drummondii*	DAV343006	Herbarium	297	53
DAV	*Juncus dubious*	DAV343031	Herbarium	170	58
DAV	*Juncus effusus subsp. pacificus*	DAV343064	Herbarium	260	71
DAV	*Juncus gerardii*	DAV343149	Herbarium	269	68
DAV	*Juncus planifolius*	DAV343253	Herbarium	261	35
DAV	*Juncus mexicanus*	DAV343277	Herbarium	213	70
DAV	*Juncus nevadensis*	DAV343320	Herbarium	161	40
E	*Juncus przewalskii*	E00044511 (ACE 601)	Herbarium	201	60
MO	*Marsippospermum grandiflorum*	J.J. Bruhl 2248	Herbarium	146	7
K	*Marsippospermum gracile*	K000352013	ENA*: ERR10977950	240	13
K	*Distichia muscoides*	K001275909	ENA*: ERR10978485	203	3
K	*Patosia clandestina*	K001393179	ENA*: ERR10977951	223	1
K	*Prionium serratum*	Kew 1997 (ERR4180160)	ENA*: ERR4180160	231	6
KRAM	*Juncus beringensis*	KRAM 7610	Herbarium	168	20
MEL	*Juncus sandwithii*	MEL 2460301	Herbarium	265	57
MEL	*Juncus revolutus*	MEL2012170	Herbarium	236	52
MO	*Juncus marginatus*	MO W.D. Stevens 39362	Herbarium	346	49
MW	*Juncus decipiens*	MW 0036772	Herbarium	150	71
MW	*Juncus jaxarticus*	MW 0812594	Herbarium	150	68
	Luzula elegans	ERR1278117	ENA*: ERR1278117	320	61
PRA	*Juncus allioides*	PRA B. Dickoré 8201	Herbarium	254	61
PRA	*Juncus imbricatus*	PRA C. Köbele 10/30/2002	Herbarium	292	71
PRA	*Distichia muscoides*	PRA C. Köbele 10/30/2002	Herbarium	259	65
PRA	*Juncus sikkimensis*	PRA G. and S Miehe 00-125-04II	Herbarium	289	59
PRA	*Juncus minimus*	PRA G. and S Miehe 00-373-02	Herbarium	256	61
PRA	*Juncus alpinoarticulatus*	PRA J. Kirschner 2/15/2005	Herbarium	252	50
PRA	*Juncus subulatus*	PRA J. Kirschner 6/2/2002	Herbarium	260	71
PRA	*Juncus castaneus*	PRA L. Klime_ 4089 (04-11-16)	Herbarium	212	61
PRA	*Juncus himalensis*	PRA L. Klime_ 4494 (04-33-10)	Herbarium	162	61
PRA	*Juncus stygius*	PRA L. Záveská Drábková 7/28/2000	Herbarium	24	5
PRA	*Juncus trifidus*	PRA L. Záveská Drábková 8/1/2000	Herbarium	174	51
PRA	*Juncus conglomeratus*	PRA L. Záveská Drábková 8/15/2000	Herbarium	260	69
PRA	*Juncus squarrosus*	PRA L. Záveská Drábková 8/15/2000	Herbarium	261	40
PRA	*Juncus triglumis*	PRA L. Záveská Drábková 8/4/2000	Herbarium	6	0
DAV	*Juncus hemiendytus*	RK 2021-19	Silica dried	212	61
DAV	*Juncus hemiendytus*	RK 2021-19	Silica dried	196	61

(Continues)

UNCORRECTED PROOFS

Herbarium code	Taxon	Voucher	Source: Accession	353 loci	Plastid loci
DAV	*Juncus hemiendytus*	RK 2021-23	Silica dried	212	61
DAV	*Juncus hemiendytus*	RK 2021-23	Silica dried	196	61
DAV	*Juncus uncialis*	RK 2023-01	Silica dried	118	2
DAV	*Juncus macrandrus*	RK 2023-03	Silica dried	238	57
DAV	*Juncus macrandrus*	RK 2023-16	Silica dried	239	58
DAV	*Juncus bryoides*	RK 2023-17	Silica dried	135	44
DAV	*Juncus saximontanus*	RK 2023-18	Silica dried	224	58
DAV	*Juncus capitatus*	RK 2021-28	Silica dried	72	52
TROM	*Luzula confusa*	TROM_V_38388	ENA*: ERR8268608	NA	66
TROM	*Luzula nivalis*	TROM_V_56647	ENA*: ERR5529777	NA	65
TROM	*Luzula parviflora*	TROM_V_651551	ENA*: ERR5554954	NA	66
TROM	*Luzula campestris*	TROM_V_66732	ENA*: ERR5529684	NA	66
TROM	*Luzula sudetica*	TROM_V_89470	ENA*: ERR5554826	NA	67
TROM	*Luzula arcuata*	TROM_V_89524	ENA*: ERR5554852	NA	61
TROM	*Luzula wahlenbergii*	TROM_V_960590	ENA*: ERR5554894	NA	66
TROM	*Luzula pilosa*	TROM_V_962165	ENA*: ERR5554983	NA	67
TROM	*Luzula spicata*	TROM_V_962459	ENA*: ERR5554743	NA	65
TROM	*Luzula sylvatica*	TROM_V_964251	ENA*: ERR5554782	NA	67
TROM	*Luzula luzuloides*	TROM_V_964549	ENA*: ERR5555243	NA	66
TROM	*Luzula multiflora*	TROM_V_968048	ENA*: ERR5555376	NA	66

*European Nucleotide Archive

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Appendix 2. Taxonomic treatment of genera in Juncaceae.

Here we propose new circumscription of *Juncinella* as a monotypic genus with *Juncinella* as its only member. We also propose a new circumscription of *Boreojuncus* to include all the North American species that were formerly included in *Juncinella*, and a new genus *Australojuncinella* that includes the South African annual species formerly included in *Juncinella*. As *Juncinella capitata* is the type specimen for the genus *Juncinella*, it is appropriate that the genus name remains with the monotypic genus that includes *Juncinella capitata*.

Key to the genera of Juncaceae (modified from Proćków and Záveská-Drábková, 2023)

1a Auricles lacerate…… *Oreojuncus*

1b Auricles absent or entire……2

2a Leaf margin hairy, at least sparsely near the sheath opening; seeds 3…… *Luzula*

2b Leaf margin not developed (leaves round, glabrous), or glabrous; seeds many……3

3a Leaf margin minutely serrulate…… *Patosia*

3b Leaf margin not developed (leaves round) or smooth……4

4a Flowers in inflorescence; anthers connective, not mucronate (if flowers occasionally solitary, then tepals shorter than 10 mm)……5

4b Flowers solitary and anthers mucronate (if anthers not mucronate, then outer tepals at least 15 mm long) ……13

5a Inflorescence racemose, a pair of floral bracteoles absent, flowers usually in heads or clusters, rarely borne ± singly……6

5b Inflorescence cymose, a pair of floral bracteoles present below each flower, flowers usually borne singly or in loose groups…… *Agathryon*

6a Leaves terete, stem-like, pungent, basal, not septate, vascular bundles scattered over most of the transversal section, lower bract apparently forming a prolongation of stem…… *Juncus*

6b Leaves flat, compressed, canaliculate, or if terete, then usually septate, not stem-like, basal or cauline, vascular

bundles usually in a subepidermal position, lower bract usually not in a position of stem prolongation…… …… …… 7

7a Leaves unitubulose or pluritubulose, perfectly septate; if pluritubulose and imperfectly septate then terete, flattened or laterally compressed…… …… *Verojuncus*

7b Leaves flat or with raised margins, not septate, or bitubulose and septate; if unitubulose and perfectly septate then anthers distinctly exserted…… ……8

8a Annuals; inflorescence bracts leaf-like, exceeding inflorescence, outer tepals much longer than inner, acuminate…… …… *Juncinella*

8b Annuals or perennials but not as above…… ……9

9a Anthers distinctly to at least partly exserted from the perianth in later stages of flowering…… …… …… *Alpinojuncus*

9b Anthers not exserted…… ……10

10a Seeds with two distinct appendages, in Sino-Himalayan region…… *Alpinojuncus*

10b Seeds without appendages, if appendages present, then in western North America……11

11a Plants perennial with 1–6 (or more) cauline leaves, if without cauline leaves then annual and occurring in western North America. Mainly in northern hemisphere. If cauline rosette is present, then *B. repens*…… …… *Boreojuncus*

11b Plants without cauline leaves, if cauline leaves present, then *Australojuncus cyperoides*. Native only in southern hemisphere…… ……. 12

12a Plants perennial…… ……*Australojuncus*

12b Plants annual…… ……*Australojuncinella*

13a Plants cushion-forming; upper part of stem densely covered with leaves; flower lateral (subterminal, axillary) …… ……14

13b Plants not cushion-forming, upper part of stem leafless, flower terminal……15

14a Leaves regularly distichous; gynophore developed, elongating during capsule ripening…… …… …… *Distichia*

14b Leaves ± spirally arranged; gynophore absent…… …… *Oxychloë*

15a Flower bracts 2, the lower one herbaceous and conspicuously longer than perianth, upper bract ± equalling perianth; capsule suborbicular to obovoid, obtuse, to c. 5 mm long, seeds without conspicuous appendages…… …… ……13. *Rostkovia*

15b Flower bracts 1 or 2, membranous, much shorter than perianth; capsule oblong to ellipsoidal, trigonous, acuminate, at least 7 mm long, seeds with two distinct appendages…… ……14. *Marsippospermum*

Juncinella (Fourr. ex V.I. Krecz. & Gontsch.) Záveská Drábková & Proćków, Mol. Phylogenet. Evol. 177 (107588): 20 (2022)

Monotypic genus with one species, *Juncinella capitata*.

Juncinella capitata (Weigel) Záveská Drábková & Proćków

Boreojuncus Záveská Drábková & Proćków, Mol. Phylogenet. Evol. 177 (107588): 21 (2022)

Perennials to small dwarf annuals, 1–15(–30) cm tall. Perennials, caespitose or rhizomatous; stems terete to slightly compressed, with leaf blade flat and not differentiated dorsiventrally, without septa. Annuals, leaves basal, canaliculate or rarely ± flat or ± terete, nonseptate, acute; auricles absent, marginal; sclerenchyma absent. Lower bract inconspicuous or leaf-like. Inflorescence terminal in annuals, usually with1–7 subsessile to pedunculate 1–12 (–20)-flowered heads; in perennials, inflorescences compound to decompound and anthelate with 1 to many flower heads, and individual flowers in the axil of a floral bract; floral bracteoles absent; peduncles not incrassate. Seeds subglobose to oblong-ellipsoid, minutely apiculate, reticulate to ± smooth, with or without appendages.

Boreojuncus bryoides (F.J. Herm.) Kenny, **comb. nov.** ≡ *Juncinella bryoides* (F.J. Herm.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus bryoides* F.J. Herm., Leafl. w. bot. 5: 117 (1948)

Boreojuncus capillaris (F.J. Herm.) Kenny, **comb. nov**. ≡ *Juncinella capillaris* (F.J. Herm.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus capillaris* F.J. Herm., Leafl. w. bot. 5: 116 (1948)

Boreojuncus digitatus (C.W. Witham & Zika) Kenny, **comb. nov.** ≡ *Juncinella digitata* (C.W. Witham & Zika) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus digitatus* C.W. Witham & Zika, J. bot. res. Inst. Texas 2(2): 775 (–781; 3 figs.) (2008)

Boreojuncus hemiendytus (F.J. Herm.) Kenny, comb. nov. ≡ *Juncinella hemiendyta* (F.J.Herm.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus hemiendytus* F.J.Herm., Leafl. W. Bot. 5: 118 (1948)

Boreojuncus hemiendytus* var. *abjectus (F.J. Herm.) Kenny, comb. nov. ≡ *Juncinella hemiendyta* var. *abjecta* (Herm.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus hemiendytus* var. *abjectus* (Herm.) Ertter, Mem. New York Bot. Gard. 39: 76 (1986) ≡ *Juncus abjectus* F.J. Herm., Leafl. W. Bot. 5: 120 (1948)

Boreojuncus kelloggii (Engelm.) Kenny, comb. nov. ≡ *Juncinella kelloggii* (Engelm.) Záveská Drábková & Proćków,

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Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus kelloggii* Engelm., Trans. Acad. Sci. St. Louis 2: 494 (1868)

Boreojuncus leiospermus (F.J.Herm.) Kenny, **comb. nov.** ≡ *Juncinella leiosperma* (F.J.Herm.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus leiospermus* F.J.Herm., Leafl. w. bot. 5: 113 (1948)

Boreojuncus leiospermus var. ***ahartii*** (Ertter) Kenny, **comb. nov.** ≡ *Juncinella leiosperma* var. *arhartii* (Ertter) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus leiospermus* var. *ahartii* Ertter, Mem. new York bot. Gard. 39: 49 (1986)

Boreojuncus luciensis (Ertter) Kenny, **comb. nov.** ≡ *Juncinella luciensis* (Ertter) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus luciensis* Ertter, Mem. New York Bot. Gard. 39: 58 (1986)

Boreojuncus tiehmii (Ertter) Kenny, comb. nov. ≡ *Juncinella tiehmii* (Ertter) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus tiehmii* Ertter, Mem. New York Bot. Gard. 39: 60 (1986)

Boreojuncus triformis (Ertter) Kenny, comb. nov. ≡ *Juncinella triformis* (Engelm.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus triformis* Engelm., Trans. Acad. Sci. St. Louis 2: 492 (1868)

Boreojuncus uncialis (Greene) Kenny, comb. nov. ≡ *Juncinella uncialis* (Greene) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus uncialis* Greene, Pittonia 2: 105 (1890)

Australojuncinella **Kenny gen. nov.**

Annuals, dwarf to small, 1–15(–30) cm tall. Leaves basal, flat or canaliculate, nonseptate, filiform; auricles absent, marginal; sclerenchyma absent. Lower bract inconspicuous. Inflorescence terminal, usually with 1–7 subsessile to pedunculate 1–12(–20)-flowered heads; floral bracteoles absent; peduncles not incrassate, stamens 6. Seeds subglobose to oblong-ellipsoid, minutely apiculate, reticulate to ± smooth; appendages absent.

Australojuncinella is distinguished by its annual habit from *Australojuncus*. Both genera are distinguished from other genera of Juncaceae by having flat or canaliculate leaves, lacking a leaf-like inflorescence bract, anthers not exserted, seed appendages absent, and lacking cauline leaves (except for *Australojuncus cyperoides*).

Australojuncinella cephalotes (Thunb.) Kenny comb. nov. ≡ *Juncinella cephalotes* (Thunb.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus cephalotes* Thunb., Prodr. Pl. Cap. 66 (1794).

Australojuncinella obliqua (Adamson) Kenny comb. nov. ≡ *Juncinella obliqua* (Adamson) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus obliquus* Adamson, J. S. African Bot. 3: 165 (1937)

Australojuncinella picta (Steud.) Kenny comb. nov. ≡ *Juncinella picta* (Steud.) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus pictus* Steud., syn. Pl. Glumac. 2: 305 (1855)

Australojuncinella rupestric (Kunth) Kenny comb. nov. ≡ *Juncinella rupestris* (Kunth) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus rupestris* Kunth, Enum. Pl. 3: 344 (1841)

Astralojuncinella scabriuscula (Kunth) Kenny comb. nov. ≡ *Juncinella scabriuscula* (Kunth) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus scabriusculus* Kunth, Enum. Pl. 3: 354 (1841)

Australojuncinella stenopetala (Adamson) Kenny comb. nov. ≡ *Juncinella stenopetala* (Adamson) Záveská Drábková & Proćków, Phytotaxa 622 (1): 017–041 (2023) ≡ *Juncus stenopetalus* Adamson, J. S. African Bot. 8: 273 (1942)

UNCORRECTED PROOFS